

Ian Leung

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EDUCATION

University of Waterloo

Expected 2029

Bachelor of Computer Science, Finance, & Statistics - Triple Major

GPA: 3.8/4.0

- President's Scholarship of Distinction (**\$5000**), Undergraduate Achievement Award (**\$1500**)
- Relevant Coursework: Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Data Analytics

WORK EXPERIENCE

Roblox

Jun 2026 – Present

Software Engineer Intern (Network Reliability) | Go, Python

San Mateo, CA

- Automated the Network Engineer push process from scratch, pushing configs to **5,000+** devices in global datacenters and simplifying deployment from a several-hour process to a **single GitHub Actions trigger**
- Orchestrated CI/CD and Temporal workflows for safe, gated deployment of configuration changes, validating changes on canary devices before progressive fleet-wide rollout with **30+** parallel workers
- Integrated **LLM-based analysis** of commit diffs, ensuring changes are safe to deploy and preventing misconfigurations/outages that could negatively impact 100M+ monthly active users
- Identified and fixed a resource leak in an internal network service, preventing API lockout and recovering **7%** CPU utilization across devices

WQC

Jan 2026 – Present

Core Developer (DevOps) | Rust, Python, AWS

Waterloo, ON

- Engineered portal.waterlooquantclub.com, a full-stack portal hosted on AWS for **700+** members, supporting JWT authentication, custom application forms, resume uploads, reviewer workflows, and Excel exports
- Solo developed grafana.waterlooquantclub.com, a Grafana observability dashboard using PostgreSQL, TLS, and AWS, providing real-time monitoring of **100+** metrics, and automatic **Discord bot alerts** for critical service failures
- Built and deployed waterlooquantclub.com/games/2v2bluff/, a real-time multiplayer **Rust/WebSocket + React** card game supporting four concurrent players, synchronized game state, and automated client/server CI/CD

Sunnybrook Research Institute

Jul 2025 – Aug 2025

Software Research Intern | Python

Toronto, ON

- Built an event tracing and playback tool to simulate medical professionals' interactions with **Focused Ultrasound** software, enabling realistic testing and diagnosis of software defects, unintended user behaviours, and edge cases
- Created an automated software testing framework which generates synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Made a logging service to validate test results against baselines, catching 4 software defects before production release

SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash

Technologies Temporal, gRPC, REST, PyTorch, Pandas, Grafana, PostgreSQL, MongoDB, React, Actions

Tools Linux, Git, Docker, AWS (EC2, S3, Lambda, etc.), OpenRouter, Jupyter, Supabase, Unity, Android Studio

PROJECTS

Pharmagotchi - Pharmacy Companion App | Android Studio, Figma, Kotlin, XML, OpenRouter

- Built a Tamagotchi-inspired **Android** health-companion app to track medications and health, backed by an offline Room database and WorkManager background reminders
- Integrated an **OpenRouter**-powered chat assistant with Retrieval-Augmented Generation (RAG) that parses health conditions and medications into structured **JSON** vitals and provides context-aware medication guidance
- Connected Fitbit tracking devices through the Google Health API, centralizing health data for seamless cross-device monitoring

PeerAssist - Peer Review Platform | Java, MongoDB, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for 20+ students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with **Apache PDFBox** and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin